

Supplementary Information III

Supplementary Methods III

DNA extraction and genomic analyses of propionate and butyrate oxidisers

Literature investigation identified three microorganisms described as syntrophic propionate oxidisers (*Syntrophobacter wolinii*, *Syntrophobacter sulfatireducens* and *Syntrophobacter pfennigii*; references 1–3) and three described as syntrophic butyrate oxidisers (*Syntrophomonas buswellii*, *Syntrophomonas erecta* and *Syntrophomonas curvata*; references 4–6) for which no genome sequences were publicly available. Three microorganisms were purchased as active cultures and subjected to DNA extraction, while purified DNA was purchased directly for the remaining three.

DNA extraction was performed according to the instructions provided with the RiboPure-Yeast kit (Invitrogen, Thermo Fisher Scientific, Waltham, MA, USA). For each extract, NanoDrop 2000 and Qubit fluorometers (Thermo Fisher Scientific, Waltham, MA, USA) were used to assess DNA purity and concentration, respectively. DNA from all six microorganisms was sent to the DiBio Sequencing Service (Department of Biology, University of Padova, Italy). Libraries were prepared using the Nextera DNA Flex Library Prep Kit (Illumina, San Diego, CA, USA) and sequenced on the Illumina NovaSeq platform using 2×150 -bp paired-end reads.

Bioinformatic analyses were performed to reconstruct the genomes of the syntrophic propionate and butyrate oxidisers. Raw-read quality was assessed using FastQC v0.11.8 and reads were trimmed using Trimmomatic. Reads were assembled using SPAdes v3.13.0, contigs shorter than 1,000 bp were removed and assembly quality was assessed using QUAST v5.0.2 [reference to be added]. Genome size and other assembly characteristics suggested the presence of contaminating sequences, which was further supported by taxonomic analysis. The datasets were therefore treated as metagenomic rather than isolate samples and the target genomes were recovered using a binning strategy. Reads were aligned to the assemblies using Bowtie2 v2.4.5 and contigs were binned using MetaBAT2 v2.2.15. Genome completeness and contamination were assessed using CheckM2 v0.1.3 (reference 7), and taxonomic classification was performed using GTDB-Tk v2.3.2 (reference 8). The reconstructed genomes were added to genomes retrieved from the literature to generate the positive classes of the corresponding function-specific training sets.

Refinement of negative classes for function-specific training sets

For functions sharing genomic components with related metabolic processes, genomes representing biologically relevant alternative functions were explicitly added to the negative class. This allowed the models to distinguish the target functions from related processes that could otherwise produce false-positive predictions. For the homoacetogenesis model, methanogens, sulfate reducers and syntrophic propionate- and butyrate-oxidising microorganisms were included as negatives because these organisms can encode components of the Wood–Ljungdahl pathway that are also present in homoacetogens. For starch degradation, sulfate reducers and syntrophic propionate- and butyrate-oxidising microorganisms were included because these anaerobic digestion-associated groups were not represented in the original MICROPHERRET background (reference 9) and could otherwise generate false-positive predictions. Similarly, hydrogen-consuming methanogens, sulfate reducers and homoacetogens were included as negatives for H_2 production to provide biologically relevant H_2 -consuming counterparts that were insufficiently represented in the original background. These genomes were incorporated after the KO-based refinement step described in the main Methods.

References

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Supplementary Figures III

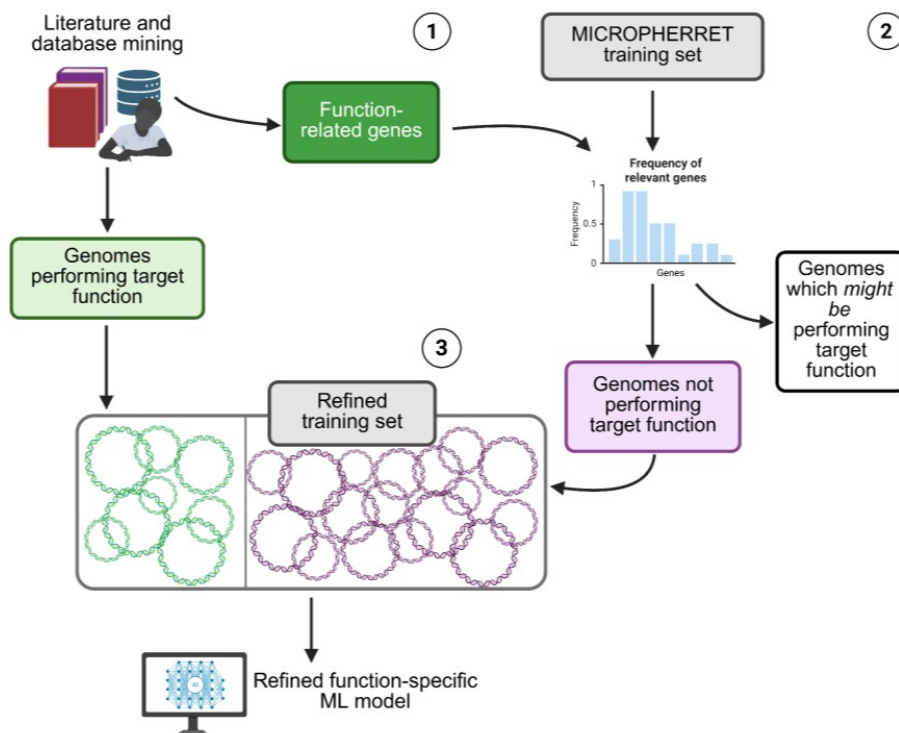


Figure III-S1. Workflow used to generate each refined function-specific training set from the original MICROPERRET dataset. Literature and databases were searched to retrieve genomes associated with each target function and the corresponding function-related genes. These genes were used to identify genomes in the MICROPERRET training set that were unlikely to perform the target function. The newly identified genomes and literature-derived positive representatives were then used to construct the refined negative and positive classes, respectively.

[Figure III-S2 to be inserted]

Figure III-S2. Pathway-level analysis of highly ranked features in the propionate- and butyrate-oxidation models. Propanoate metabolism was identified for the propionate-oxidation model, whereas butanoate metabolism was identified for the butyrate-oxidation model.

[Figure III-S3 to be inserted]

Figure III-S3. Numbers of MAGs predicted to perform each of the nine anaerobic digestion functions. Cellulose and starch degradation were predicted in substantially more genomes than the other functions.

[Figure III-S4 to be inserted]

Figure III-S4. Predicted relative abundance of the nine functional populations across feedstock categories. Differences among feedstocks were evaluated using Kruskal–Wallis tests.

[Figure III-S5 to be inserted]

Figure III-S5. Overlap between predicted volatile fatty acid oxidisers and sulfate reducers. Five of 15 predicted propionate-oxidising MAGs and three of 21 predicted butyrate-oxidising MAGs were also predicted as sulfate reducers.

Supplementary Tables III

Table III-S1. Composition of the positive classes used for the nine function-specific models. Positive classes contained 14–51 genomes depending on the availability of experimentally or literature-supported representatives.

[Table III-S1 to be inserted]

Table III-S2. Selected algorithm and hyperparameter configuration for each function-specific model. Selected models were subsequently refitted using the complete function-specific training sets.

[Table III-S2 to be inserted]

Table III-S3. Feedstock classifications used in the comparative analyses.

[Table III-S3 to be inserted]

Table III-S4. Characteristics of the anaerobic digestion studies and datasets included in the applications.

[Table III-S4 to be inserted]